

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q24-2485

Farrell 138 kV
December 29, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2485. The Interconnection Customer proposes a 90.0 MW solar photovoltaic facility interconnecting at the Farrell 138 kV substation in the Virginia area. The request was received on December 08, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 424 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,409,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Van Tyle – Cholame 138 KV: 131.5% loading (220 MVA) following the outage of the Lewisburg – Siloam Springs 138 KV line; DFAX = 8.3%.

Fumarole – Riverton 138 KV: 126.9% loading (209 MVA) following the outage of the Farrell – Depere 138 KV; DFAX = 11.0%.

For the loss of International Paper Valliant Ok – Kearneyville 345 KV, the Depere – Farrell 138 KV line loads to 126.9% of its 301 MVA rating. Project DFAX: 26.5%.

For the loss of International Paper Valliant Ok – Kearneyville 345 KV, the Rochester Hydro – Depere 138 KV Line loads to 126.6% of its 397 MVA rating. Project DFAX: 21.4%.

For the loss of Kearneyville – International Paper Valliant Ok 345 KV, the Rochester Hydro – Cholame 138 KV loads to 124.9% of its 352 MVA rating. Project DFAX: 16.8%.

Siloam Springs – Cholame 138 KV: 124.0% loading (239 MVA) following the outage of the Van Tyle Substation – Cholame Substation 138 KV; DFAX = 13.2%.

The Twin Lakes Alte – Wpcp 138 KV is overloaded to 117.2% of its 273 MVA emergency rating for the loss of the Twin Lakes Alte Substation – Huntley 138 KV. The project contributes 5.3% DFAX (4.8 MW) to this overload.

Old Chapel – Belleair 138 KV Line: 116.8% loading (403 MVA) following the outage of the Santa Rosa Energy Ctr Substation – Balch 138 KV Line; DFAX = 6.7%.

For the loss of Santa Rosa Energy Ctr – Balch Substation 138 KV, the Hp Hood Ct – Springfield 138 KV line loads to 113.2% of its 448 MVA rating. Project DFAX: 6.7%.

The Fumarole – Depere 138 KV is overloaded to 110.0% of its 208 MVA emergency rating for the loss of the Depere – Farrell 138 KV. The project contributes 11.0% DFAX (9.9 MW) to this overload.

Double Toll Gate – Abbotsford 138 KV line: 104.9% loading (330 MVA) following the outage of the Double Toll Gate – Springfield Substation 138 KV line; DFAX = 7.2%.

The Old Chapel – Hp Hood Ct 138 KV is overloaded to 101.7% of its 477 MVA emergency rating for the loss of the Balch – Santa Rosa Energy Center 138 KV L. The project contributes 6.7% DFAX (6.0 MW) to this overload.

The Columbus City 345/1 kV XFMR is overloaded to 100.9% of its 755 MVA emergency rating for the loss of the Kearneyville – Clark County 345 KV. The project contributes 9.2% DFAX (8.2 MW) to this overload.

The Rverton 230 – Van Tyle 138 KV is overloaded to 100.8% of its 344 MVA emergency rating for the loss of the Siloam Springs – Lewisburg 138 KV. The project contributes 5.1% DFAX (4.6 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Van Tyle – Cholame 138 KV	Rebuild line with higher-capacity conductor	\$19,625,000	28% (\$5,529,000)
Riverton – Fumarole 138 KV	Rebuild line with higher-capacity conductor	\$7,985,000	9% (\$743,000)
Depere – Farrell 138 KV line	Rebuild line with higher-capacity conductor	\$4,564,000	14% (\$625,000)
Rochester Hydro – Dpere 138 KV Line	Rebuild line with higher-capacity conductor	\$15,710,000	9% (\$1,461,000)
Rochester Hydro – Cholame 138 KV	Rebuild line with higher-capacity conductor	\$5,925,000	9% (\$559,000)
Siloam Springs – Cholame 138 KV	Rebuild line with higher-capacity conductor	\$851,000	28% (\$240,000)
Twin Lakes Alte – Wpcp 138 KV	Rebuild line with higher-capacity conductor	\$26,169,000	14% (\$3,580,000)
Old Chapel – Belleair 138 KV Line	Rebuild line with higher-capacity conductor	\$828,000	28% (\$234,000)
Hp Hood Ct – Springfield 138 KV line	Rebuild line with higher-capacity conductor	\$4,128,000	28% (\$1,166,000)
Depere – Fumarole 138 KV Line	Reconductor line	\$4,111,000	9% (\$383,000)
Double Toll Gate – Abbotsford 138 KV	Reconductor line	\$2,334,000	7% (\$172,000)
Old Chapel – Hp Hood Ct 138 KV	Reconductor line	\$3,280,000	28% (\$926,000)
Columbus City 345/1 kV Autotransformer	Replace transformer with higher-rated unit	\$15,029,000	3% (\$454,000)
Rverton 230 – Van Tyle 138 KV	Reconductor line	\$1,197,000	28% (\$337,000)
	Total allocated to project		\$16,409,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.